Review

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Environment-Aware Multi-Sensor Fusion for Maritime Domain Awareness: A Comprehensive Review

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Abstract: Environmental conditions fundamentally shape the performance, reliability, and interpretability of modern maritime sensing systems. This review synthesizes current knowledge on how wind, significant wave height, swell, atmospheric variability, and radiative properties influence Synthetic Aperture Radar (SAR) backscatter statistics, wake and shadow visibility, optical image contrast, and the stability of Automatic Identification System (AIS) signals. These environmental factors induce non-stationary ocean clutter, distort ship-related signatures, and generate environmentally driven AIS delays or darkening events, thereby amplifying uncertainty throughout the multi-sensor fusion process. This review discusses that existing multi-sensor fusion frameworks, including SAR–AIS integration, while operationally effective, remain structurally limited when environmental context is not explicitly incorporated. In particular, SAR, optical, AIS, and Radio Frequency (RF) sensors exhibit distinct environment-dependent failure modes under varying sea states and atmospheric conditions, which can propagate into false detections, imbalanced sensor confidence, and reduced interpretability at the fusion stage. Building on recent advances in machine learning–based detection, high-revisit SAR constellations, and environment-informed detectability modeling, we argue that future Maritime Domain Awareness (MDA) systems must adopt environmental information as a central organizing principle for multi-sensor fusion rather than as a peripheral auxiliary input. This environment-aware multi-sensor fusion perspective is especially relevant for the Korean Exclusive Economic Zone (EEZ) and broader East Asian waters, where shallow-water turbidity-driven clutter, monsoon-induced environmental variability, swell intrusions, and dense vessel traffic create particularly challenging sensing conditions. The integration of high-resolution SAR, high-revisit satellite constellations, satellite AIS, RF sensors, and MetOcean datasets enables the development of environment-conditioned surveillance frameworks tailored to such region-specific environments. Overall, this review positions environment-aware multi-sensor fusion–based MDA not as an optional enhancement but as a structural requirement for next-generation maritime surveillance, providing a roadmap for both research and operational development in dynamically evolving maritime regions.

Keywords: Environment-aware maritime surveillance, MDA, Sea-state effects on vessel detection, High-clutter maritime environments, Multi-modal sensor fusion, EEZ

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1. Introduction

Over the past two decades, maritime security has been increasingly shaped by illicit and high-risk activities—including Illegal, Unreported and Unregulated (IUU) fishing, deliberate Automatic Identification System (AIS) switch-off, ship-to-ship transfers, and transnational smuggling—that frequently occur beyond the reach of coastal radars and patrol assets (Li et al., 2025a; Wang et al.,

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2024a). Satellite-based sensing using Synthetic Aperture Radar (SAR), optical systems, satellite AIS, and Radio Frequency (RF) geolocation has therefore become indispensable for modern Maritime Domain Awareness (MDA) (Mehta et al., 2025; Javed et al., 2024). Although recent advances in spatial resolution, revisit frequency, and Deep Learning (DL)-based ship detection have markedly expanded the operational capabilities of satellite surveillance, current systems and much of the research community remain fundamentally sensor-centric (Mehta et al., 2025; Yasir et al., 2024b). Environmental factors are often addressed only superficially, treated as incidental sources of degradation rather than as variables that fundamentally shape what can be detected —and how reliably it can be detected (Crisp, 2004). In practice, dismissing environmental variability introduces systematic vulnerabilities. High sea states, strong winds, long-period swell, persistent cloud fields, and atmospheric ducting can all reduce detectability, distort key signatures such as wakes and shadows, or degrade AIS and RF reception (Pleskachevsky et al., 2019a; Pleskachevsky et al., 2021). These impacts manifest not only as missed detections but also as increases in false alarms and ambiguous or low-confidence outputs. Consequently, an effective MDA system must recognize environmental forcing as a first-order determinant of satellite sensing performance, rather than a peripheral constraint.

Remote sensing of the ocean is governed by the interaction between electromagnetic waves and a dynamically evolving sea surface characterized by gravity–capillary waves, longer-period swell, and a complex slope distribution. These physical processes control fundamental parameters—including sea clutter statistics, normalized radar cross section, and target-to-clutter ratio—that directly impact both detectability and detection reliability. Even under identical imaging geometry, variations in wind and wave conditions can lead to markedly different outcomes in SAR ship detection and wake interpretation (Bezerra et al., 2023; Tings et al., 2019a). Optical sensing is likewise strongly constrained by environmental variability. Clouds, fog, aerosols, and sun glint can substantially reduce image visibility and contrast, frequently masking wakes or small vessels entirely. AIS and RF-based systems are similarly affected. Sea-induced vessel motion, such as rolling and pitching, can alter antenna orientation and reduce effective signal reception, while atmospheric conditions influence Very High Frequency (VHF) propagation by modifying attenuation characteristics and transmission range. In congested maritime corridors, message collisions further lead to irregular or missing AIS reports, which may be erroneously interpreted as deliberate dark behavior (Rizaev et al., 2022; Tings et al., 2019b). Collectively, these examples demonstrate that environmental conditions are not merely ancillary noise sources but constitute primary drivers of uncertainty that affect detection performance and reliability across all satellite sensing modalities.

Despite the recognized importance of environmental effects, existing maritime surveillance research has largely evolved in a fragmented manner, focusing on individual sensors or algorithmic components rather than on their coupled interaction with environmental forcing. In particular, vessel detection research has been dominated by SAR-based studies, reflecting the practical advantage of SAR's all-weather and day–night imaging capability. As a result, a substantial body of literature has been established around SAR-centered detection problems. Seminal work by Crisp and subsequent studies have systematically reviewed Constant False Alarm Rate (CFAR)-based and feature-based ship detection algorithms, sea clutter modeling, and associated operational challenges in SAR imagery (Crisp, 2004; Tello et al., 2006). In parallel, extensive research has been conducted on sea-state and wind retrieval from SAR, with multiple studies summarizing empirical and semi-empirical approaches for estimating wind speed and sea-state parameters from C-band and X-band observations (Pleskachevsky et al., 2019a; Pleskachevsky and Tings, 2019). Ship wake detection has also been the subject of dedicated reviews, encompassing both SAR- and optical-based techniques, including traditional image-processing methods and more recent learning-based approaches (Mazzeo et al., 2024).

More recently, large-scale SAR ship detection datasets and deep learning techniques have been introduced and benchmarked for maritime scenes (Humayun et al., 2023; Yasir et al., 2024b), alongside studies on multi-sensor fusion strategies that exploit the complementary characteristics of SAR, optical imagery, and AIS for dark ship detection and traffic validation (Li et al., 2025a; Mourampetzis, 2025). However, while these studies provide valuable insights into individual sensing modalities and algorithmic advances, they predominantly organize knowledge along sensor-specific or method-specific lines. As a consequence, the systematic role of environmental conditions in shaping not only vessel detectability but also detection reliability and interpretability has received comparatively limited attention. This gap motivates the need for a unified, environment-aware perspective that explicitly links environmental forcing, sensor response, and surveillance performance.

To our knowledge, there is no review that systematically organizes maritime security sensing from an environment-aware

Table 1. Summary of the core functional domains of MDA (European Maritime Safety Agency, 2023; Galdelli et al., 2021; Li et al., 2025a; Paolo et al., 2022)

MDA Functional Domain	Operational Objective	Primary Targets	Key Satellite Sensors
Dark Vessel Detection	Identify vessels operating without AIS or with irregular transmissions	Non-cooperative ships, AIS gaps	SAR, Optical, S-AIS, RF
IUU Fishing Monitoring	Detect and characterize illegal, unreported, and unregulated fishing activities	Fishing vessels, abnormal motion patterns	SAR, Optical, AIS
Ship-to-Ship (STS) Transfer Monitoring	Identify close-proximity vessel interactions indicative of illicit transfer	Paired vessels, prolonged co-location	SAR, Optical, AIS
Marine Pollution Surveillance	Detect oil spills, chemical discharges, and debris	Oil slicks, anomalous surface films	SAR, Optical
Maritime Traffic Monitoring	Track vessel density and movement patterns	Commercial shipping lanes, ports	AIS, SAR, Optical
Search and Rescue (SAR)	Support the detection of distressed vessels or debris	Small boats, life rafts, wreckage	SAR, Optical
Border and EEZ Surveillance	Enforce sovereignty and maritime security	Unauthorized entry, loitering vessels	SAR, AIS, RF

perspective, connecting physical scattering, sea state retrieval, detection algorithms, and multi-sensor fusion under the unifying question: "How do environmental conditions control what satellites can see, and how should MDA systems adapt?" Furthermore, existing MDA-oriented works rarely consider region-specific environmental regimes, such as those of the Korean Exclusive Economic Zone (EEZ), where monsoon winds, strong currents, and complex coastal morphology combine to create highly variable detection conditions (World Meteorological Organization, 1998; Pleskachevsky et al., 2019b).

This review addresses these gaps through four contributions:

- Synthesizing SAR, optical, AIS, and RF sensing from an explicitly environment-aware perspective, describing how wind, wave height, swell, and atmospheric conditions shape detectability and reliability.
- Introducing a structured detection-chain framework—environment → surface/atmospheric state → sensor signal and clutter → detectability and reliability → operational decision-making—and reorganizing prior work within this architecture.
- Identifying critical gaps in current research, including the need for environment-labeled multi-sensor datasets, adaptive detection pipelines, and dynamic fusion approaches driven by MetOcean priors.
- Discussing operational implications for the Korean EEZ, emphasizing how monsoon variability, tides, and complex bathymetry produce variable detection regimes and how environment-aware approaches can be operationalized using existing Korean satellite and MetOcean systems.

Through this approach, the review aims to move beyond a purely technical summary and provide both a conceptual and operational foundation for environment-aware MDA that explicitly incorporates detection reliability. Table 1 presents a summary of the core functional domains of MDA.

2. Materials and Methods

This section offers an expanded synthesis of the physics linking marine environmental states to sensing responses. Unlike typical sensor-centered descriptions, this section emphasizes the bidirectional coupling between oceanographic processes and remote sensing observables, providing the physical basis for the environment-aware MDA framework developed in later sections.

2.1. Wind–Wave–Surface Roughness Processes

Wind stress transferred from the atmospheric boundary layer generates a continuous spectrum of ocean waves, including capillary waves ($\lambda < 1$ cm), gravity–capillary waves (1–10 cm), and longer gravity waves (> 10 cm). These components coexist and interact, and the resulting surface roughness and spectral characteristics are strongly governed by wind speed, fetch, and atmospheric stability. In the microwave domain, radar backscatter is primarily controlled by short gravity–capillary waves that satisfy the Bragg resonance condition, $2k_s=k_r$. According to classical scattering theory summarized by Valenzuela (1978), these resonant short waves dominate the normalized radar cross section (σ^0), meaning that wind-driven surface roughness is the principal determinant of brightness variations in SAR imagery (Elyouncha,

Table 2. Implications of wind-speed regimes for sea-surface characteristics, SAR backscattering, and ship/wake detectability (Plant, 2002; Rosenberg et al., 2019; Schwendeman and Thomson, 2015; Tings et al., 2019a; Valenzuela, 1978)

Wind Speed	Sea-Surface Characteristics	SAR Backscatter & Detectability Implications
< 2 m/s	Very smooth surface	Low σ^0, high vessel contrast
2–8 m/s	Developed gravity–capillary waves	Stable clutter, robust detection of ships and wakes
> 10–12 m/s	Breaking waves, strong roughness	Increased σ^0 variance, reduced detectability for small vessels

2020; Wenguang et al., 2008).

As wind intensity increases, the sea surface transitions from a smooth, specular reflector to a rough, diffuse-scattering background. Under low winds, Bragg-resonant waves remain weak, σ^0 is low, and vessel contrast is high. With increasing wind speed, σ^0 rises quasi-linearly, and sea clutter becomes more spatially variable. Under strong winds, wave breaking and non-Bragg scattering generate heavy-tailed clutter statistics, significantly reducing the detectability of small or low-Radar Cross Section (RCS) vessels (Pleskachevsky et al., 2019a; Wenguang et al., 2008). These environmental regimes exhibit distinct sea-surface scattering characteristics and detection behaviors from an operational perspective and can be systematically classified according to wind-speed ranges. Table 2 summarizes the representative wind-speed regimes together with their associated sea-surface characteristics, SAR backscattering behavior, and implications for ship and wake detectability.

The wind–wave interaction affects not only the mean level of σ^0 but also the statistical form of sea clutter distributions. Under strong winds, clutter evolves from near-Gaussian behavior to heavy-tailed forms such as K-distributions or compound models, reflecting an elevated probability of extreme backscatter spikes (Pai and Kiang, 2024; Wenguang et al., 2008). This variability implies that vessel detectability is not an intrinsic property of the target but is strongly conditioned by the prevailing sea state, underscoring the necessity of environment-aware detection and fusion frameworks in maritime surveillance.

2.2. Swell, Wave Height, and Long-Wave Modulation

Long-period wave components such as swell and significant wave height do not directly generate Bragg-scale scattering, but they strongly modulate short gravity–capillary waves through tilt, hydrodynamic, and velocity-bunching mechanisms. Classical theory and recent reviews consistently describe this process as a low-frequency modulation of short waves that controls SAR backscatter. Long waves periodically increase and decrease local incidence angle and short-wave roughness on their front and rear slopes, producing alternating bright–dark bands in SAR intensity (Centre for Remote Imaging, Sensing and Processing, 2001; Ouchi and Yoshida, 2023). These patterns have been observed in imagery from multiple spaceborne sensors, including SEASAT, European Remote Sensing Satellite (ERS), RADARSAT, and Sentinel-1. Internal waves and swell concentrate or disperse short waves, and several case studies using ERS-2 and RADARSAT show kilometer-scale parallel bright–dark stripes that form quasi-linear features on the ocean surface (Kim et al., 2007). In a single scene, these bands may coexist with fronts, slicks, or ship-generated internal waves. Table 3 shows the expected physical effects of SAR in dynamic marine environments and their implications for ship detection.

In the context of ship wakes, recent performance analyses and reviews show that Kelvin wakes, turbulent wakes, and ship-generated internal waves all appear as linear bright–dark features whose visibility depends strongly on local sea state such as wind, wave height, and wind direction (Mazzeo et al., 2024). Under rough conditions, the slope distribution broadens and breaking waves and whitecaps increase, resulting in both Bragg and non-Bragg scattering, as well as azimuth streaks that populate the image with linear patterns (Ouchi and Yoshida, 2023). When bright–dark bands generated by long waves or internal waves overlap with Kelvin patterns, turbulent wakes, or low-backscatter regions near ship shadows, they occupy similar directional and

Table 3. Expected SAR effects under different sea states and implications for ship detection Hs denotes the significant wave height (Hasselmann et al., 1985; Sun et al., 2018; Tings et al., 2019a).

Sea-State Condition	Expected SAR Effect	Implication for Detection
Long-period swell, moderate Hs	Azimuthal streaks / backscatter modulation	Wake-like artifacts → false alarms
High Hs with breaking waves	Increased σ^0 variance, bright speckle, non-Bragg scattering	Clutter dominates → small-vessel miss, low contrast
Calm / low swell & low Hs	Smooth background, low clutter	High vessel contrast, reliable detection

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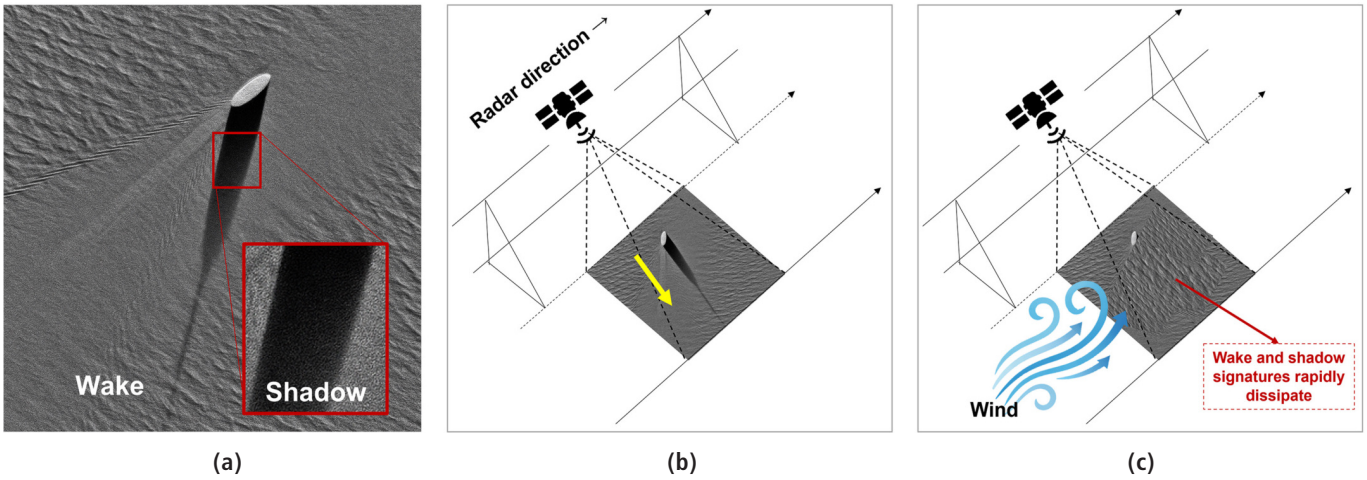

Fig. 1. Examples of ship wake and shadow patterns: (a) conceptual illustration of wake and shadow, (b) example of ship shadow orientation and geometry, and (c) changes in wake and shadow patterns under strong wind conditions.

spatial scales. This increases the number of candidates produced by Radon/Hough-based line detection and highlights the need for auxiliary sea-state information to separate environmental streaks from genuine wake or shadow signatures (Mazzeo et al., 2024; Tings et al., 2019b). Fig. 1 illustrates the wake and shadow patterns of ships in SAR imagery, visually presenting their physical formation mechanisms and variations in dynamic marine environments.

2.3. Atmospheric and Radiative Modulators

Because optical sensors observe radiance transmitted through the atmosphere rather than the sea surface itself, atmospheric and radiative conditions constitute the primary constraint on the usability of marine optical imagery. Environmental factors such as clouds, fog, aerosols, sun-glint, and water-vapor absorption dynamically modify visibility, contrast, color fidelity, and band-specific signal-to-noise ratio (SNR), causing the same sensor to operate under markedly different radiometric regimes depending on observation conditions (Graziano et al., 2017; Liu et al., 2021; Mazzeo et al., 2024).

Clouds and low-level fog impose an essentially binary limitation on optical sensing. Over the global ocean, mean cloud cover approaches approximately 70%, and in many regions, as much as 70–80% of optical scenes are partially or fully obscured. Even when imagery is acquired, low-altitude fog layers preferentially attenuate shorter wavelengths, often yielding scenes in which the proportion of usable sea-surface signal is severely reduced. As a result, optical sensing may fail entirely under these conditions, regardless of sensor resolution or algorithmic sophistication.

Aerosols and haze further degrade optical performance by increasing atmospheric scattering. Elevated aerosol optical depth significantly reduces image contrast and color accuracy, particularly in the blue–green spectral bands that are critical for marine observation (Frouin et al., 2019). This degradation leads to low-contrast scenes in which small vessels, wakes, or offshore structures become difficult to distinguish from the background, even in otherwise cloud-free conditions. Sun-glint represents a geometry-driven failure mode that is particularly problematic in low-wind environments. When the sun–sea sensor geometry satisfies conditions for specular reflection, strong glint from the sea surface can enter directly into the sensor's field of view. This effect occurs as a function of solar incidence angle, wave-slope distribution, and sensor viewing geometry, and can saturate or dominate the background radiance.

Consequently, large portions of optical scenes become unusable for vessel or wake detection, as the glint signal overwhelms subtle target–background contrast. Sun-glint, therefore, constitutes a major limitation for optical maritime surveillance, especially when observation geometry cannot be adjusted or when onboard exposure control is insufficient (Liu et al., 2021; Mazzeo et al., 2024). Finally, water vapor and other gaseous absorbers modulate band-specific SNR through wavelength-dependent absorption, introducing variability in atmospheric correction accuracy and cross-band detectability. These radiative effects create heterogeneous confidence levels across spectral bands within the same scene. Because RF and VHF AIS signal propagation is also sensitive to atmospheric refractive index structure, evaporation ducts, and temperature inversions, such environmental variability provides

a physical basis for dynamically assigning sensor and band weights in subsequent optical–SAR–AIS fusion frameworks (Ward et al., 2006).

3. Satellite Sensing Systems from an Environment-Aware Viewpoint

In this section, we reinterpret satellite sensing modalities through the lens of environmental sensitivity. While each sensing system has known capabilities, the emphasis here is on how their performance envelopes contract or expand under different sea-state and atmospheric scenarios.

3.1. SAR Sensing in Dynamic Sea States

SAR is a core modality for maritime surveillance under cloud cover, adverse weather, and nighttime conditions. However, its performance is strongly conditioned by sea-state variability, frequency-band response, imaging geometry, polarization, and the statistical behavior of sea clutter (Lin et al., 2008; Pleskachevsky et al., 2019a). As sea states change, the spectral composition of backscatter, clutter distributions, and target–background contrast shift rapidly, introducing substantial variability in vessel detectability and algorithmic robustness.

3.1.1. Frequency-Dependent Sea-Surface Backscatter (σ^0) and Environmental Sensitivity

The sea-surface scattering characteristics of SAR systems vary fundamentally with radar frequency as a result of the physical interaction between radar wavelength and sea-surface roughness scales. In microwave SAR, the normalized radar cross section (σ^0) of the ocean surface is primarily governed by Bragg scattering from short surface waves that satisfy the resonance condition. The range of surface-wave wavelengths contributing to this resonance is directly determined by the radar wavelength. Consequently, X-, C-, and L-band SAR systems sample different portions of the sea-surface wave spectrum, leading to pronounced differences in σ^0 magnitude and statistical behavior under identical wind and wave conditions.

X-band SAR, characterized by its short wavelength, is highly sensitive to centimeter-scale capillary–gravity waves. Under low-wind and low-wave conditions, the amplitudes of these short waves remain weak, resulting in a low background σ^0 and a high target-to-clutter ratio. This property makes X-band SAR particularly effective for detecting small vessels or weak scatterers in calm seas. However, as wind speed increases, the density and amplitude of short-wave components rise rapidly, and wave breaking as well as non-Bragg scattering mechanisms increasingly contribute to the backscatter. These processes lead to a sharp increase in clutter variance and non-stationarity, causing a rapid degradation in detection performance under rough sea states. As a result, X-band SAR is prone to clutter amplification and elevated false-alarm rates in high-wind and high-wave environments (Vachon et al., 2013).

C-band SAR occupies an intermediate regime in which the balance between radar wavelength and dominant short-wave scales remains relatively stable. Bragg scattering continues to dominate the backscatter response under low-to-moderate wind and wave conditions, typically corresponding to sea states with significant wave heights of approximately 1–3 m and wind speeds of about 3–10 m/s, resulting in more gradual changes in both the mean backscatter level and its statistical variability. This physical behavior allows C-band SAR to maintain relatively consistent image quality across diverse sea states and provides the theoretical basis for its widespread use in global ocean observation, sea-surface wind retrieval, and monitoring of medium-to-large vessels.

Nevertheless, under strong winds and high sea states, the increasing contribution of non-Bragg scattering can still lead to elevated clutter variability and reduced detectability of small targets (Mestre-Quereda and Lopez-Sanchez, 2025). L-band SAR, with its longer wavelength, exhibits reduced sensitivity to short-scale surface roughness and is instead more influenced by longer-period gravity waves and swell-induced surface slope modulation. As a result, background clutter levels tend to increase more slowly with wind speed, and the overall scattering field remains comparatively stable even under rough sea conditions. This robustness makes L-band SAR advantageous in high sea states. However, when the radar wavelength approaches or exceeds the characteristic dimensions of small vessels, target-to-background contrast is reduced, imposing an inherent limitation on small-vessel detectability. Thus, while L-band SAR offers enhanced robustness under severe environmental conditions, it is structurally less sensitive to small maritime targets (Del Prete et al., 2023).

These frequency-dependent differences in environmental sensitivity and σ^0 behavior are not merely empirical observations but arise directly from the physical matching between radar wavelength and the sea-surface wave spectrum. Accordingly, SAR frequency selection should be guided by both the prevailing sea state and the intended surveillance objective. In environment-

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Table 4. Characteristics of X-, C-, and L-band (Del Prete et al., 2023; Mestre-Quereda and Lopez-Sanchez, 2025; Vachon et al., 2013)

Frequency Band	Environmental Sensitivity	Strengths	Limitations
X-band	High (short-wave sensitive)	Strong small-vessel detection at low winds	Severe clutter escalation at high winds
C-band	Moderate	Stable across diverse sea states	Slightly reduced sensitivity to very small targets
L-band	Low (long-wave dominant)	Robust in rough seas and swell	Poor contrast for small vessels

aware maritime surveillance frameworks, adaptive strategies that explicitly account for frequency-dependent σ^0 characteristics are therefore essential for achieving robust and reliable detection performance. Table 4 summarizes the environmental sensitivity and response strength across different SAR frequency bands, along with the corresponding limitations.

3.1.2. Geometry and Polarization

SAR imaging geometry governs the dominant scattering mechanism. Low incidence angles tend to amplify specular reflections from the sea surface, reducing vessel contrast, whereas higher incidence angles enhance diffuse and Bragg scattering, improving detectability. Polarization further modulates this interaction: VV is highly sensitive to wind-roughened surfaces, while HH and cross-pol channels provide more stable responses and capture non-Bragg scattering such as breaking waves. Empirical studies confirm that polarization-aware detection frameworks outperform single-pol systems across variable sea states (Zhang et al., 2022).

3.1.3. Sea Clutter Statistics Under Dynamic Conditions

Sea clutter is inherently non-Gaussian and often modeled with heavy-tailed distributions such as K-distributions, Weibull models, or log-normal mixtures. Critically, the shape and scale parameters of these distributions vary substantially with wind speed, wave height, and surface turbulence (Ishii et al., 2011). This non-stationarity undermines fixed-threshold detectors such as classical CFAR and highlights the necessity of adaptive thresholding, dynamic clutter maps, and sea-state-conditioned detection models. Recent work demonstrates that clutter-aware and sea-state-aware approaches provide markedly more stable performance under operational conditions (Del Prete et al., 2023).

3.2. Optical Satellite Imaging Under Environmental Constraints

Optical satellite imaging provides rich semantic information such as vessel shape, hull color, plume signatures, and fine-scale wake structures, but its usefulness is fundamentally limited by atmospheric and illumination conditions. Unlike SAR, optical systems rely on reflected solar radiation, meaning visibility is greatly affected by cloud cover, haze, aerosols, and water-vapor absorption. These factors reduce radiometric contrast between the vessel and the surrounding ocean surface, directly impacting the reliability of silhouette-based vessel detection and the ability to distinguish spectral differences between hull materials and seawater. Wake visibility is similarly dependent on illumination geometry: unfavorable sun angles, excessive sun-glint, or off-nadir viewing can mask subtle brightness variations associated with Kelvin wakes or turbulent trails, while wave shadowing in rough seas can obscure or distort wake signatures altogether (Kay et al., 2013; Voiland, 2014). Furthermore, plume detection—such as identifying exhaust trails or discharge events—requires both high radiometric sensitivity and stable atmospheric conditions. Bright plumes are readily detectable in clear-sky conditions, but even moderate haze or off-axis lighting can suppress the necessary contrast. As a result, optical imaging performance exhibits strong scene-to-scene variability, emphasizing the need for environment-aware interpretation and adaptive fusion with radar or RF sources when used for maritime surveillance.

3.3. Satellite AIS: Environment-Dependent Reliability

The reporting pattern of the AIS is significantly influenced not only by vessel intent but also by systemic and environmental constraints, including coverage structure, radio wave propagation characteristics, and traffic density. The horizontal coverage of terrestrial AIS (T-AIS), typically installed in coastal areas, is limited to tens of kilometers (approximately 15–40 nautical miles, and a maximum of 40–60 nautical miles for high-altitude receivers) due to the Earth's curvature and VHF propagation characteristics. Satellite AIS (S-AIS) serves as the primary means of reception in the offshore and open sea regions beyond this range (Jankowski et al., 2021; Kontas, 2024; Scarbrough et al., 2017). In low- to mid-latitude offshore areas, S-AIS performance is determined by the orbital period and ground revisit time, which are dictated by the Low Earth Orbit (LEO) geometry and the satellite constellation configuration. It has been reported that the interval between adjacent passes by the same satellite is typically tens of minutes

to 1–2 hours, and observational gaps on the scale of several hours can occur depending on the latitude (Eriksen et al., 2018). Consequently, even if a vessel continuously transmits AIS messages as required by regulations, the trajectories reconstructed by the receiving entity commonly exhibit non-uniform timestamps and intermittent data gaps (Jankowski et al., 2021).

Furthermore, meteorological and oceanographic conditions additionally degrade the propagation quality of the AIS VHF signal. Heavy precipitation, atmospheric noise, haze/fog, high waves, and rough sea surfaces can attenuate or scatter the VHF signal or induce multipath effects, thereby reducing the SNR of the received signal (Moraga, 2024; Wang et al., 2024c). Reports also indicate that in rough sea states, the rolling and pitching of the vessel, combined with fluctuations in the horizon, can intermittently obstruct the line-of-sight between the transmitting and receiving antennas, leading to a decrease in reception success rate (Wang et al., 2024b). Analysis suggests that in high-latitude and remote areas, the superposition of these propagation losses and satellite coverage limitations frequently results in AIS gaps and omissions unrelated to intentional blockage (Arifin et al., 2024; Jankowski et al., 2021). Traffic congestion also constitutes a significant systemic factor.

As AIS operates on a Time Division Multiple Access (TDMA) basis, numerous vessels compete for the same time slots in high-density areas such as shipping lanes, ports, and fishing grounds. This competition leads to an increase in packet collisions, partial decoding, and message losses. These congestion and collision issues are particularly pronounced in the wide field of view of S-AIS. Multiple studies have highlighted that reception loss sharply increases when vessels in different sea areas, but within the satellite's line of sight, utilize the identical time slot (Eriksen et al., 2018; Mieczyńska and Czarnowski, 2021). Finally, at the system and operational levels, cases of irregular AIS data omission are also reported due to equipment failures, antenna blockage, power and Global Positioning System (GPS) issues, and terrestrial receiver network gaps (Jankowski et al., 2021; Owen, 2025). It is repeatedly

Environmental Factors Affecting SAR-AIS Fusion

Environmental Factor	Physical Mechanism	Impact on AIS Reliability	Implications for AIS–SAR Fusion
High waves / vessel roll–pitch	Antenna tilt → reduced line-of-sight	Weak / intermittent AIS signals	Missing AIS ≠ intentional dark vessel
Rainfall / sea spray	VHF signal attenuation	Lower received signal strength	Greater uncertainty in AIS–SAR matching
Coastal multipath	Terrain/structure reflections	Distorted or delayed AIS packets	Apparent false positions, timestamp inconsistency
Evaporation ducts / elevated ducts	Refractive trapping / extended propagation	Anomalous long-range reception or dropouts	Requires **environment-aware interpretation**
High traffic density	TDMA message collisions	Packet loss, reduced update rate	**Relax spatial/temporal tolerances** in fusion
Strong winds / turbulence	Antenna vibration, platform instability	Increased signal variability	Additional noise in validation datasets

Fig. 2. Characteristics of SAR-AIS fusion affected by the environmental factors (Rautiainen et al., 2025; Wang and Zhang, 2017; Wang et al., 2018).

emphasized in recent review papers and AIS utilization research that these factors can cause reporting delays and intermittent gaps independent of intentional "dark vessel" behavior. Therefore, inferring illegal or non-cooperative activities based solely on the presence of missing data segments is cautioned against as being risky (Arifin et al., 2024). Fig. 2 illustrates the potential signal effects on AIS under varying environmental conditions.

3.4. RF Emitter Detection Under Atmospheric Variability

Satellite-based RF emitter detection systems rely on predictable electromagnetic propagation paths to estimate direction-of-arrival and emitter geolocation. However, in maritime environments, atmospheric refractivity can vary substantially in space and time, leading to significant deviations from standard propagation assumptions. Variations in temperature, humidity, and pressure —particularly within the marine atmospheric boundary layer— can give rise to pronounced refractivity gradients that fundamentally alter RF propagation behavior (Joint Chiefs of Staff, 1994; Thomas, 2006). One of the most prominent manifestations of such variability is the formation of atmospheric ducts. Under specific refractivity profiles, including evaporation ducts near the sea surface and elevated inversion layers, RF signals may become trapped within a refractive layer. This ducting effect can substantially extend propagation range beyond the radio horizon or, conversely, create shadow zones with reduced signal reception. As a result, RF signals may arrive at receiving satellites along non-geometric paths, violating straight-line propagation assumptions and increasing uncertainty in both bearing estimation and emitter localization (Huang et al., 2022).

Experimental and modeling studies have demonstrated that atmospheric ducting can strongly affect maritime radar and RF system performance. For example, analyses of navigation radar under ducting conditions show that signals can propagate well beyond the expected line-of-sight range, while standard-atmosphere assumptions lead to misinterpretation of target location and range. These effects are not anomalous edge cases but recurring phenomena in marine and coastal environments, where evaporation-driven humidity gradients and temperature inversions frequently occur. Such refractivity-induced propagation variability is particularly pronounced in coastal regions characterized by strong land–sea thermal contrasts. Around the Korean Peninsula, rapid changes in humidity, sea surface temperature, and atmospheric stability can occur on sub-daily timescales, creating highly dynamic refractive environments. In these conditions, RF detection frameworks that assume homogeneous or standard atmospheric propagation are prone to producing false bearings, mislocalized emitters, and inconsistent detection performance (Joint Chiefs of Staff, 1994; Thomas, 2006). Consequently, robust RF-based maritime surveillance requires the explicit incorporation of refractivity-aware propagation modeling and environmental priors. Accounting for atmospheric variability is essential to reduce geolocation uncertainty, improve detection consistency, and ensure reliable interpretation of RF emitter observations within multi-sensor maritime domain awareness frameworks. Fig. 3 illustrates the RF signal transmission, reception, and analysis process for ship detection.

4. Environment-Aware Vessel Detectability and Algorithmic Implications

The performance of SAR-based vessel detection is constrained not only by sensor parameters such as resolution, polarization, and incidence angle, but also by metocean factors including wind speed, significant wave height, and overall sea state. Using TerraSAR-X imagery and AIS data, Tings et al. (2019a) developed a data-driven ship detectability model that expresses the Probability of Detection (PD) as a function of vessel length, wind speed, wave height, and incidence angle. Their results show quantitatively that

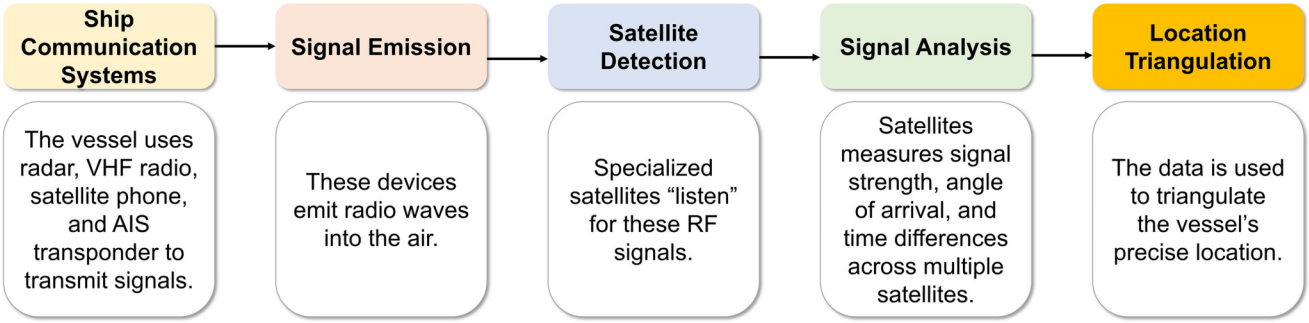

Fig. 3. Workflow of radio frequency (RF) (WINDWARD, 2025).

PD increases under low wind, low waves, and large incidence angles, whereas it decreases under high wind, high waves, and small incidence angles (Velotto and Tings, 2018). These findings confirm that vessel detectability is determined systematically by the combined interactions of sensor parameters and wind–wave–sea-state conditions (Tings et al., 2023).

4.1. Environment- and Frequency-Dependent SAR Ship Detectability

The performance of SAR-based ship detection is not determined solely by sensor resolution or algorithmic design, but is fundamentally governed by the interaction between marine environmental conditions and sensor configurations. In particular, meteorological and oceanographic (MetOcean) factors—such as wind speed, significant wave height, and overall sea state—modulate both the mean level and statistical variability of the sea-surface normalized radar cross section (σ^0). These changes directly affect the target-to-clutter ratio and, consequently, the stability and reliability of ship detection. Importantly, this environmental dependence manifests differently across SAR frequency bands, such that X-, C-, and L-band systems exhibit distinct detection characteristics even under identical sea-state conditions.

To date, many quantitative studies on ship detectability have relied primarily on high-resolution X-band SAR data, owing to its strong sensitivity to short-scale surface roughness and its pronounced response to variations in wind and wave conditions. However, the key mechanisms discussed in this section are not confined to a single frequency band. Clutter enhancement under rough seas, reductions in specular sea-surface return with increasing incidence angle, and the resulting changes in vessel–background contrast arise from physical scattering processes that are common across SAR frequencies. The magnitude and expression of these effects, however, vary with radar wavelength, leading to frequency-dependent differences in environmental sensitivity and detection limits.

From this perspective, ship detection is reframed here not as a sensor-specific performance issue, but as a detectability problem governed by the combined effects of environmental conditions and radar frequency. The following subsections examine detection behavior under low and high sea states, respectively, and analyze the coupled influence of incidence angle and sea state on ship detectability. While representative X-band studies are used to illustrate these relationships quantitatively, the discussion emphasizes the broader applicability of the underlying physical trends across SAR frequencies. Through this approach, the section highlights the role of environment-aware SAR operation and frequency selection in achieving robust maritime surveillance performance.

4.1.1. Low-Wind Conditions: Enhanced Vessel–Background Contrast

Under low-wind, low-sea-state conditions, sea-surface turbulence is weak and whitecaps are minimal, producing low and spatially homogeneous background backscatter (σ^0). Metallic vessels, therefore, appear as bright targets against a dark ocean background, resulting in enhanced vessel–background contrast and increased PD. The TerraSAR-X detectability model of Tings and Velotto (2018) shows that, for wind speeds of 2–6 m/s, low significant wave height (~0.5 m), and moderate-to-high incidence angles (20–45°), PD remains consistently high for medium and large vessels. Their study provides detectability charts specifying the minimum vessel length required to achieve PD = 0.95 as a function of wind, wave height, and incidence angle—demonstrating that the minimum detectable ship size becomes smaller under calmer conditions. In such environments, sea-clutter statistics remain relatively stable, allowing conventional Gaussian-based CFAR to perform effectively. Even when CFAR thresholds are raised conservatively, PD does not drop sharply. Both Tings et al. (2019a) and Velotto and Tings (2018) confirm qualitatively and quantitatively that calm sea states combined with sufficient incidence angles allow robust detection of small to large vessels.

4.1.2. High-Wind / High Sea-State Conditions: Increased Clutter and False Alarms

In contrast, as wind speed and wave height increase, turbulence, breaking waves, and whitecaps generate sea clutter with high variance and heavy-tailed statistics. An et al. (2018) analyzed sea clutter from Gaofen-3 SAR and showed that high-resolution marine clutter is better modeled by K-distribution, Gamma, or Rayleigh distributions rather than Gaussian statistics, with the K-distribution particularly effective for spiky sea clutter. Similarly, Wang et al. (2008) proposed an α-stable–distribution CFAR for RADARSAT-1 imagery and demonstrated experimentally that, under heavy-tailed clutter, Gaussian CFAR underestimates false alarms and exhibits rapidly degraded PD.

More recent studies have introduced composite clutter models such as K-Rayleigh or Tri-Modal Discrete texture (3-MD) distributions to better represent clutter across different incidence angles and geographic regions, enabling more adaptive CFAR designs. For example, Mungara et al. (2024) confirmed that the K-distribution

remains an effective statistical model for a wide range of real marine clutter environments. Additionally, Zhao et al. (2019) showed that CFAR performance under K-distributed clutter deteriorates severely in multi-target scenes and proposed a Density-Based Spatial Clustering of Application with Noise (DBSCAN)-CFAR approach combining Artificial Neural Network (ANN) features with density clustering, achieving more stable false-alarm control in the presence of sea spikes and heavy clutter. Across these studies, a consistent conclusion emerges: under strong wind and high sea states, sea-clutter distributions shift to non-Gaussian, heavy-tailed behavior, invalidating the homogeneous-Gaussian assumptions of classical CFAR. In such environments, lowering thresholds detects both small vessels and whitecaps simultaneously —causing false alarms to surge—while raising thresholds results in missed detections of actual vessels. As Velotto and Tings (2018) documented, retaining small-vessel detection under high wind requires accepting higher false-alarm rates, whereas reducing false alarms inevitably causes small targets to be lost.

4.1.3. Coupled Incidence-Angle × Sea-State Models

The combined effects of incidence angle and sea state are most clearly summarized in the TerraSAR-X detectability model of Tings et al. (2019a). Using thousands of AIS–SAR matched samples, they constructed a logistic regression model of PD = f (wind, wave height, incidence angle, vessel length) and visualized the results as two-dimensional detectability charts over wind-incidence-angle space. The results reveal a set of intuitive yet quantitatively confirmed patterns, indicating that detection probability improves under lower wind speeds, lower significant wave heights, higher incidence angles, and for larger vessels.

In high-resolution TerraSAR-X imagery, incidence-angle effects are particularly pronounced. Under identical sea-state conditions, vessel–background contrast is noticeably weaker at low incidence angles of approximately 20°, whereas it becomes significantly enhanced at higher incidence angles near 40° (Tings et al., 2019a). Velotto and Tings (2018) further demonstrated that, for small vessels, even high-resolution SAR imaging modes can exhibit unstable detection performance when incidence angles are suboptimal or wind conditions are strong. In such cases, relatively small variations in sea state may determine whether a vessel signature remains above or falls below the CFAR detection threshold. These observations can be interpreted through a combined empirical and theoretical perspective.

As the incidence angle increases, the contribution of specular reflection from the sea surface is reduced, leading to a relative enhancement of vessel contrast against the background. Under low-wind and low-wave conditions combined with moderate-to-high incidence angles, SAR imagery typically yields high probabilities of detection with low false-alarm rates. Conversely, high wind and wave conditions coupled with low incidence angles amplify sea clutter through increased specular and non-Bragg scattering, resulting in elevated false-alarm rates and a higher likelihood of missed detections. The detectability charts introduced by Tings et al. (2019a), which relate minimum detectable vessel length to a target probability of detection (PD = 0.95) under varying metocean and geometric conditions, provide an operationally valuable reference. These charts support informed decisions on SAR imaging mode selection, incidence-angle planning, and threshold presetting within environment-aware maritime surveillance frameworks.

Although the coupled effects of incidence angle and sea state are most clearly observed in X-band SAR, their physical origin is rooted in sea-surface scattering mechanisms that are not inherently frequency-specific. For C-band SAR, studies likewise report the same directional trends, namely an enhancement of background clutter with increasing wind speed and wave height, and an improvement in vessel–background contrast with increasing incidence angle. However, owing to the longer radar wavelength, these variations tend to occur more gradually than in X-band systems. In the case of L-band SAR, the reduced sensitivity to short-scale surface roughness limits clutter amplification even under strong wind and high sea-state conditions, but this robustness comes at the cost of structurally reduced sensitivity to small vessels. Consequently, while the strength and manifestation of environment-dependent detectability vary across SAR frequency bands, the fundamental interaction between sea state, incidence angle, and vessel detectability remains consistent.

4.2. Challenges in Wake and Shadow Analysis

Vessel wakes and radar shadows remain foundational cues in SAR-based maritime surveillance, facilitating precise ship localization, heading and speed estimation, and identification of dark or non-cooperative vessels. However, real-world sea states frequently give rise to environmental patterns—such as long-period swell, wind streaks, internal waves, and high waves—that generate quasi-linear or shadow-like features. These environmental artifacts can mimic vessel-induced signals or disrupt shadow continuity, undermining detection reliability and increasing false alarms. Below, we review two primary challenge classes and highlight evidence and mitigation directions from existing literature.

Several studies have reported that environmentally generated linear or quasi-linear structures frequently appear in SAR imagery, independent of any vessel activity. Internal waves produce convergence–divergence zones that manifest as alternating bright–dark stripes extending over several kilometers. Klemas (2012) summarized that such internal-wave fronts are clearly observed in RADARSAT and ERS SAR scenes as coherent, elongated patterns. Zhang et al. (2025) further demonstrated that internal-wave bands form continuous stripe-like signatures in SAR imagery using automated segmentation experiments. Wind streaks are another well-documented source of linear patterns. Zhao et al. (2016) showed that wind streaks appear as elongated, directionally organized surface features in satellite SAR observations. These structures are non-wake in origin but form distinct linear patterns in amplitude imagery.

Wake-detection algorithms that rely on Radon or Hough transforms are designed to enhance linear or elongated features, and thus may also respond to non-wake structures. Yang et al. (2020) noted that elongated non-wake signatures can be emphasized in Radon-domain processing, and proposed Bayesian post-processing steps to reduce their influence. Recent deep-learning studies report similar behavior. Ding et al. (2023), using Gaofen-3 SAR imagery, showed that internal-wave bands, coastal boundaries, and other elongated surface features can be mistakenly highlighted as wake-like patterns during automatic detection. These findings indicate that non-wake linear features may be detected as wake candidates by modern learning-based approaches.

Radar shadows—defined as low-backscatter regions formed behind or adjacent to a vessel—are widely used in ship–shadow detection algorithms and for validating vessel geometry in SAR imagery. The radiometric visibility of such shadows has been shown to depend strongly on the roughness of the surrounding sea surface. As surface roughness increases, local contrast between illuminated and shadowed regions is reduced, which in turn degrades the extraction of shadow-based features. Studies focusing on shadow segmentation and texture analysis consistently acknowledge that sea-surface roughness modulates the contrast structure of radar shadows, thereby influencing their detectability and interpretability. Experimental observations reported in the literature indicate that shadow contrast varies systematically with sea-surface roughness and that this variability directly affects the difficulty of shadow-based analysis.

Beyond shadow-specific effects, a broad range of environmental processes—including internal waves, wind streaks, and other surface manifestations of ocean–atmosphere interaction—generate distinct linear or quasi-linear structures in SAR imagery. These features are often elongated, spatially coherent, and may exhibit strong radiometric contrast, making them visually similar to vessel wakes. As a result, traditional wake-detection approaches based on Radon or Hough transforms may respond to such non-wake linear features. Recent deep-learning–based models have likewise been shown to emphasize internal-wave bands, wind-induced streaks, or coastal boundaries as wake-like patterns, particularly when trained on datasets lacking explicit environmental labeling. Taken together, these findings demonstrate that high sea states characterized by overlapping swell systems, internal

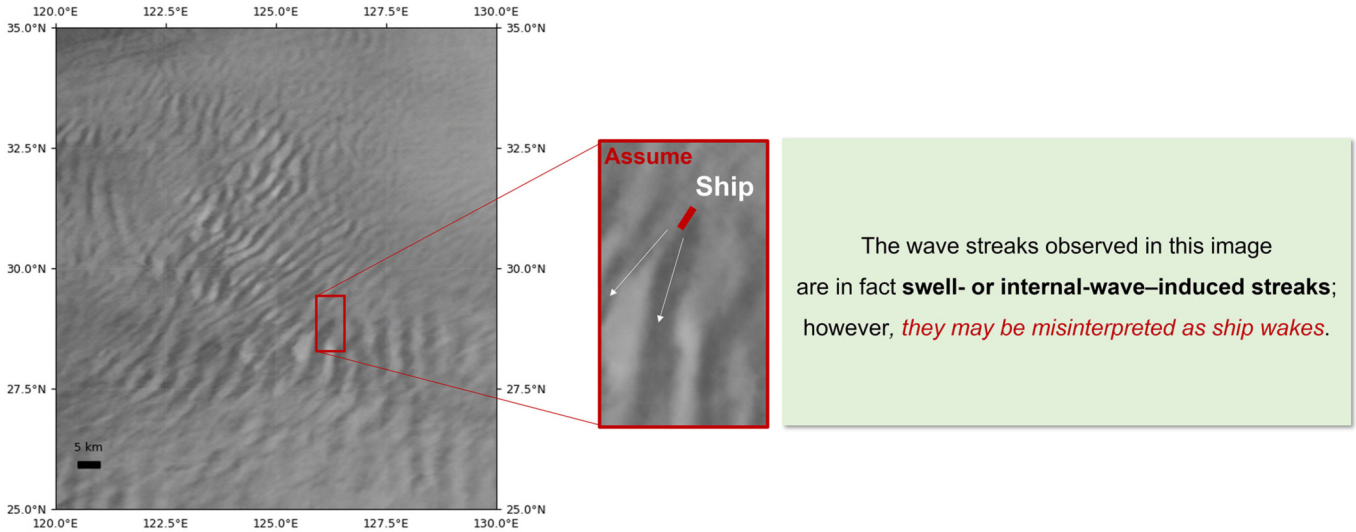

Fig. 4. Example of surface-wave (or internal-wave) signatures in SAR imagery. Strong patterns of short or long waves on the surface may be misinterpreted as ship wakes (Mazzeo et al., 2024; Tings et al., 2019b).

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waves, and wind streaks pose persistent challenges for both wake and shadow analysis. In such conditions, environmentally generated streaks frequently share geometric and textural properties with true vessel-induced wakes, thereby undermining the reliability of both classical linear-feature detectors and modern deep-learning–based approaches. Fig. 4 presents an example of streak patterns caused by internal waves (swell), which exhibit patterns similar to ship wake streaks and may therefore lead to false detections of ship presence in wake-based ship detection.

4.3. DL Techniques for SAR Ship Detection in Dynamic Environments

DL-based SAR ship detectors, utilizing structures such as Convolutional Neural Networks (CNNs), attention modules, and Transformers, have demonstrated significant performance improvements over traditional CFAR methods in terms of both accuracy and flexibility, as reported in numerous studies. Li et al. (2022) summarized 177 deep learning studies on SAR ship detection using public datasets such as the SAR Ship Detection Dataset (SSDD) and the High-Resolution SAR Images Dataset (HRSID). Their quantitative comparison confirmed the general superiority of deep learning detectors—including Faster Region-based CNN (R-CNN), You Only Look Once (YOLO), and Single Shot Detector (SSD) variants—over classical CFAR-based algorithms across precision, recall, and mean Average Precision (mAP).

However, recent studies also highlight a critical limitation: the actual maritime environment is highly variable in space and time, leading to a persistent domain shift between the sea states present in the training data and the conditions encountered during operational deployment. According to the review by Li et al. (2022), while SSDD is a representative public dataset comprising 1,160 SAR images and 2,456 ship bounding boxes, it primarily offers only image-label pairs acquired under specific sensor and sea conditions, crucially lacking environmental metadata such as wind speed, wave height, or sea state. This suggests that most deep learning models are trained on "image-label pairs without explicit environmental conditions," structurally restricting their generalization capability under extreme sea states (e.g., high waves, strong winds, severe swell, or low-quality/low-SNR images).

To mitigate this limitation, recent research has explored the use of domain adaptation and generative model-based data augmentation to reduce the statistical drift due to varying sea states. For instance, Pu et al. (2023) treat the detection challenge in low-quality and high-interference SAR images as a domain shift problem. They utilized Cycle-Consistent Generative Adversarial Network (CycleGAN)-based image-to-image translation to synthesize "high-interference-style SAR images," which were then incorporated into the detector training process. The resulting performance gains—an mAP improvement of 11.0% on GaoFen-3 and 3.22% on A High-Resolution SAR Rotation Ship Detection Dataset (SRSDD)-V1.0 datasets—demonstrated that generative domain adaptation can substantially enhance both the detection performance and the robustness of models in high-noise and high-clutter environments. Numerous studies have also reported the direct use of generative models for data augmentation. Zou et al. (2020) demonstrated that generating multi-scale, high-resolution SAR ship images using Multiscale Wasserstein Auxiliary Classifier GAN (MW-ACGAN) and combining them with YOLOv3 resulted in an mAP improvement of up to 1.01% on the HRSID dataset, suggesting a solution for sample scarcity in high-resolution SAR. Furthermore, Kwon et al. (2022) proposed a method to improve the performance of deep learning ship detectors in low-sample environments using GAN-based synthetic SAR data. Ju and Hu (2022) introduced a GAN-based "visual filtering" network called FilterGAN to generate noise-suppressed pre-processed images on the AIR-SARShip-1.0 dataset, showing a significant enhancement in the subsequent ship detection performance.

From a more fundamental perspective, Lou et al. (2022) proposed a generative knowledge transfer framework to mitigate feature disparities between heterogeneous domains and consequently reduce missed detection and false alarm rates. Moreover, the review by Li et al. (2022) suggests future research directions, including "designing detection algorithms specialized for SAR image characteristics, improving performance on small ships, lightweighting for real-time detection, and establishing large-scale standardized datasets by integrating several smaller ones." This review also indirectly suggests the future need for public benchmarks that include sea-state metadata like wind speed, wave height, and incidence angle. In summary, while current deep learning-based SAR ship detection shows superior performance compared to traditional CFAR-based techniques, a fundamental constraint persists: environmental and noise conditions not reflected in the training data cause domain shift during operational deployment. Recent studies on domain adaptation, GAN-based data augmentation, and generative knowledge transfer represent initial steps toward mitigating this statistical drift. Ultimately, designing a learning pipeline that explicitly considers sea-state conditions and developing datasets that include environmental metadata are presented as essential next-step challenges.

5. Multi-Sensor Fusion in Environment-Aware Framework

Multi-sensor fusion has become essential for maritime surveillance, particularly in scenarios involving dark vessels, AIS spoofing, small craft embedded in complex sea clutter, and operations in polar or adverse-weather regions. Early comprehensive reviews—such as Zhao et al. (2014)—organized the landscape of spaceborne SAR–AIS integrated monitoring, identifying key challenges in data association, ship identification, and multi-sensor fusion strategies. More recently, commercial and operational maritime platforms have adopted the fusion of SAR, optical, RF, and AIS streams to build a coherent maritime situational picture, indicating that research concepts are transitioning into operational architectures. The rapid growth of Artificial Intelligence (AI)-based multimodal analytics has further accelerated this trend, leveraging fused AIS, radar, imagery, and RF data for maritime security, illegal fishing detection, and vessel behavior monitoring. Table 5 presents a comparison between single-sensor and multi-sensor detection in MDA and discusses the necessity of multi-sensor fusion.

5.1. Environment-Informed SAR–AIS Fusion

5.1.1. Foundations of SAR–AIS Data Fusion

The foundational framework for SAR–AIS fusion is presented in the seminal review by Zhao et al. (2014), which surveys techniques for integrating SAR detections with spaceborne and terrestrial AIS by addressing geometric misalignment, timing discrepancies, ship heading/size constraints, and multi-attribute matching strategies. Later studies, such as Rodger and Guida (2020), extended this concept by incorporating SAR-derived ship classification and size priors into AIS association, demonstrating improved fusion robustness in dense shipping regions. Korean coastal studies using KOMPSAT-5 SAR and AIS further validated the feasibility of operational SAR–AIS fused monitoring in semi-enclosed seas with heavy marine traffic (Kim et al., 2020a; Kim et al., 2018). While these works highlight complementary strengths of SAR and AIS, most early studies treat metocean conditions only implicitly—as background clutter or noise—without directly modeling their influence on detection or data association (Hou et al., 2021; Velotto, 2022).

Table 5. Methods and characteristics of single-sensor and multi-sensor detection studies, and the necessity of sensor fusion (Achiri et al., 2018; Galdelli et al., 2021; Graziano et al., 2019; Guerriero et al., 2008; Kim and Jung, 2017; Li et al., 2025a; Talpur et al., 2025; Yasir et al., 2024a; Zhao et al., 2014)

MDA Functional Domain	Single-Sensor Approaches	Multi-Sensor Approaches	Added Value of Fusion
General vessel detection	• SAR-only ship detection • CFAR / feature-based methods • DL on SAR imagery • Performance limited by clutter & geometry	• SAR or optical + AIS association • Image-based detection + identity matching	• False-alarm reduction • Improved track continuity • Detection–identity consistency
Dark vessel detection	• AIS-only gap analysis • Trajectory anomalies • Cannot separate intent vs. reception loss	• SAR/optical detections + AIS absence • Cross-validation of non-reporting vessels	• Discrimination of intentional dark behavior • Reduced misinterpretation of AIS gaps
Wake and track analysis	• SAR/optical wake detection only • Geometric/spectral / DL methods • Confusion with swell & wind streaks	• Wake candidates validated by AIS heading/speed • Joint wake–vessel association	• Suppression of environment-induced false wakes • Improved wake continuity assessment
Ship-to-ship transfer (STS)	• AIS proximity & loitering analysis • Ineffective for AIS-off cases	• AIS-based candidate detection • SAR/optical confirmation of vessel proximity	• Detection of AIS-dark STS events • Visual confirmation of interactions
Pollution & oil-spill monitoring	• SAR dark-spot detection only • Frequent look-alike false alarms	• SAR + wind/wave/current data • Optional optical confirmation	• Reduced ambiguity • Improved spill–look-alike discrimination
AIS reliability & traffic analysis	• AIS message statistics only • Collision & density analysis	• AIS vs. SAR/optical vessel counts • Gap and omission assessment	• Diagnosis of AIS coverage limits • Confidence-aware traffic monitoring

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5.1.2. Quantifying Environmental Impacts on SAR Detectability

Independent research streams have quantified how sea state affects SAR-based ship detection. Tings et al. (2019a) proposed a statistical ship detectability model for TerraSAR-X, expressing the probability of detection as a function of wind speed, significant wave height, incidence angle, and vessel length. Their results quantitatively demonstrated a clear relationship between sea state and detectability, showing that detectability decreases markedly for small vessels as wind speed and wave height increase and as incidence angle decreases. Complementarily, Bezerra et al. (2023) conducted large-scale CFAR detection experiments using Sentinel-1 and demonstrated that wave age causes systematic variations in the background clutter distribution, significantly shifting false-alarm rates across young wind-sea, old wind-sea, and swell conditions. European Space Agency (ESA) tutorials similarly emphasize that variations in wind speed, wave height, and directional sea state cause non-stationary sea clutter statistics, complicating CFAR-based detection (Colin et al., 2022; Velotto, 2022). In combination, these studies reveal that environmental variability is a core determinant of SAR detectability and false-alarm behavior, yet it has not been fully incorporated into SAR–AIS fusion frameworks.

5.1.3. Initial Attempts toward Environment-Aware Fusion

Although few studies explicitly integrate environmental models into SAR–AIS matching, several works incorporate external contextual layers into multi-sensor fusion. Guerriero et al. (2008) introduced a Bayesian anomaly detection model combining radar, AIS, and SAR to optimize surveillance coverage and SAR tasking by representing sensor detection capabilities probabilistically. Kazemi et al. (2013) proposed and evaluated an open data-based approach for maritime surveillance anomaly detection. The authors introduced a framework for complementarily fusing external open data, such as weather conditions, port information, and regulatory mandates, with traditional data sources like existing AIS reports. They assert that the effective detection of genuinely anomalous vessel behavior is only possible by incorporating contextual factors, including environmental conditions, vessel routing, and operational regulations.

More recent multimodal DL approaches incorporate AIS and optical context into SAR ship recognition networks (Ziling et al., 2024). However, these studies generally aim to improve detection accuracy rather than explicitly integrating quantitative metocean–detectability models (Tings et al., 2019a; Bezerra et al., 2023) into the SAR–AIS fusion pipeline. Thus, the literature suggests three parallel but not yet unified directions: (1) SAR–AIS fusion, (2) metocean-aware SAR detection modeling, and (3) contextual anomaly detection frameworks—leaving fully environment-informed SAR–AIS fusion as an open research frontier.

5.2. SAR–Optical Fusion with Sea-State Constraints

5.2.1. Evolution of SAR–Optical Bimodal Fusion

SAR–optical fusion has been one of the most active areas in multimodal ship detection. Ahmed et al. (2025) reviewed SAR–optical dual-modality methods, demonstrating accuracy gains over single-modality approaches. Zhang et al. (2026) proposed a multimodal hybrid fusion framework addressing resolution and geometric inconsistencies between SAR and optical image pairs, achieving improved robustness in complex coastal scenes.

5.2.2. Relevance to Dark Vessels and AIS Inconsistencies

SAR–optical–AIS collaboration has become a dominant strategy for identifying dark vessels—those that deactivate or falsify AIS signals. Li et al. (2025a) demonstrated a collaborative framework that uses SAR to provide all-weather detections and optical imagery for high-resolution shape confirmation, combined with AIS multi-feature association for dark vessel screening. Mourampetzis (2025) similarly concluded that high-resolution optical imagery is indispensable for validating SAR detections in illegal ship-to-ship (STS) transfer investigations. These studies implicitly follow a trigger–confirm paradigm, where SAR provides robust initial cues and optical imagery performs refined visual confirmation—reflecting complementary sensor characteristics.

5.2.3 Environmental Constraints and Tasking Considerations

Explicit modeling of sea-state constraints within SAR–optical fusion remains limited. Nevertheless, environmental limitations are well-characterized for each modality. Optical sensing is strongly constrained by cloud cover, fog, aerosols, and sun glint, motivating extensive research on cloud-aware optical tasking. On the SAR side, wave-age-dependent clutter statistics significantly affect CFAR performance, as demonstrated by Bezerra et al. (2023). Thus, while SAR–optical fusion is mature in detection, classification, and dark-vessel analysis, integration of explicit sea-state or metocean constraints into fusion or tasking strategies remains at an early stage in the literature. Fig. 5 illustrates the optical–SAR fusion framework proposed by Ahmed et al. (2025)

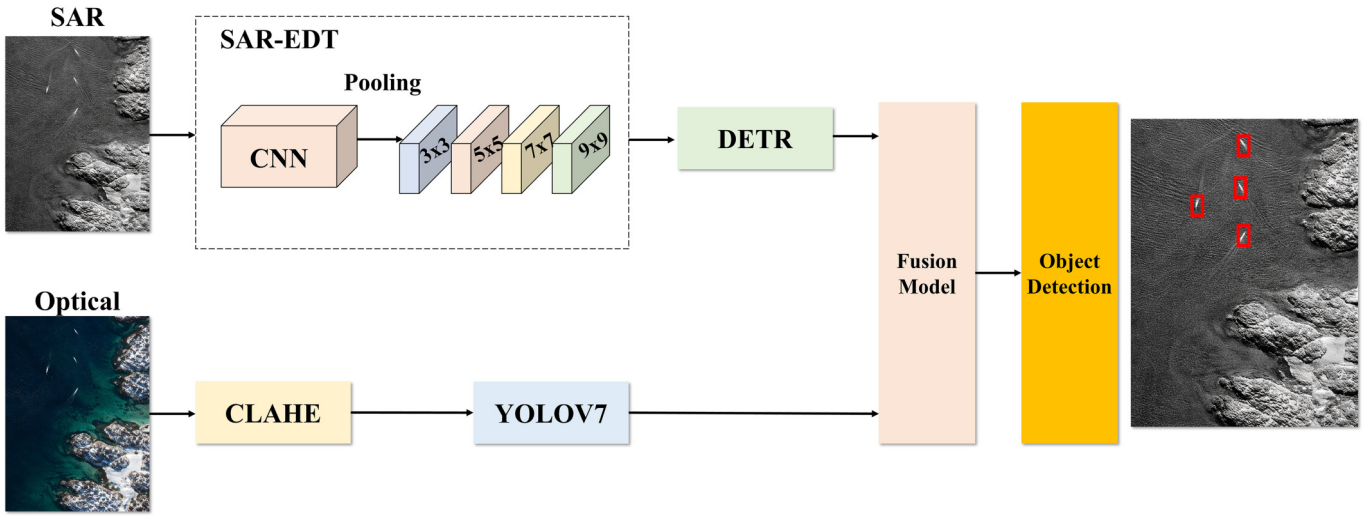

Fig. 5. Example of a flowchart of proposed ship detection in SAR and optical imagery. DETR: Detection Transformer, CLAHE: Contrast Limited Adaptive Histogram Equalization (Ahmed et al., 2025).

5.3. RF–SAR–AIS Triple Fusion

5.3.1. RF–AIS Fusion and Dark-Vessel Detection

Satellite-based RF sensing has recently emerged as a powerful complement to AIS, particularly for detecting non-cooperative or AIS-disabled vessels. Yang and Chowdhury (2025) analyzed multi-band RF detections (3 GHz, 9.4 GHz) in the Yellow Sea and Korea Strait and showed that RF emissions can reveal vessel presence where AIS signals are missing or manipulated. Commercial systems such as Unseenlabs further demonstrate operational capability by locating vessels from RF emissions independent of AIS or optical visibility (European Space Agency, 2024).

5.3.2. Emerging RF–SAR–AIS Triple-Fusion Studies

Triple fusion of RF, SAR, and AIS is still an emerging area, but has gained attention in difficult maritime domains such as Arctic waters. Sørensen et al. (2024) documented how RF provides emitter presence, SAR provides high-resolution structure and sea-ice context, and AIS supplies cooperative navigation data—collectively improving surveillance robustness. Earlier work by Guerriero et al. (2008) proposed a model for multi-sensor data fusion—integrating Radar, AIS, and SAR data—and Bayesian anomaly detection, which quantifies the value of sensor data for maritime surveillance performance and presents an approach

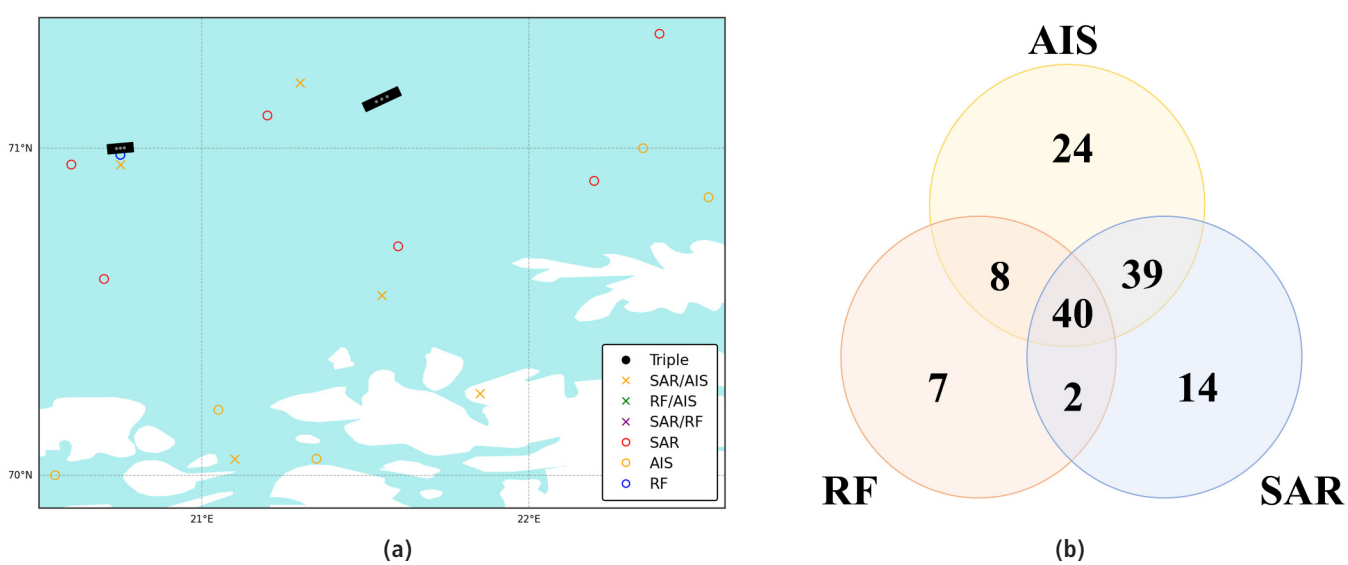

Fig. 6. Example of RF/AIS/SAR fusion. (a) Results of the RF/AIS/SAR fusion in the Arctic region. (b) Triple, double, and single matches across the dataset (Sørensen et al., 2024).

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to SAR tasking for optimal surveillance effectiveness (Carthel et al., 2007). Kazemi et al. (2013) focused on the framework and necessity of integrating open data sources for effective anomaly detection in maritime surveillance.

5.3.3. Advantages in Adverse-Weather and Low-Visibility Environments

Since RF and SAR are electromagnetic sensors resilient to cloud, fog, and nighttime conditions, RF–SAR–AIS integration provides high continuity where optical sensors fail. Recent security-oriented studies and commercial platforms highlight the combined use of RF emissions with SAR detections and AIS behavior profiles to track vessels engaging in AIS spoofing or illicit fishing under severe weather or low-light conditions (Collecte Localisation Satellites, 2024; WINDWARD, 2025). Overall, the literature indicates that RF–SAR–AIS fusion enhances detection of non-cooperative vessels, supports anomaly detection, and provides operational resilience in degraded visual environments. Fig. 6 illustrates the results and analysis of RF/AIS/SAR triple-fusion detection in the Arctic Ocean reported by Sørensen et al. (2024).

5.4. AI-Based Multi-Modal Fusion

5.4.1. Multi-Source Remote Sensing Fusion for Ship Recognition

AI-driven fusion of SAR, optical, multispectral, and radar imagery has grown substantially. Li et al. (2025b) summarize pixel-, feature-, and decision-level fusion approaches, consistently reporting performance gains relative to single-sensor systems. Zhang et al. (2021) proposed a ship target recognition algorithm based on Bayesian inference and evidence theory, demonstrating that the recognition performance is significantly improved when fusing ship features obtained from various sensors compared to using a single sensor.

5.4.2. Deep-Learning and Graph-Based AIS–Imagery Radar Fusion

Fusion architectures incorporating AIS have also expanded. Pu et al. (2025) proposed TrackAISNet, which fuses AIS time series with trajectory imagery to achieve higher ship-type classification accuracy than either modality alone. Wang et al. (2025) proposed MSTrack, a vessel detection and tracking framework that fuses multi-source sensor data, including maritime radar, visible light cameras (RGB), and AIS. This research overcame the limitations of single sensors and demonstrated a significant improvement in detection accuracy and tracking consistency for small vessels in complex maritime environments, proving that multi-modal data fusion is essential for maritime surveillance and safety. Research applying Graph Neural Networks (GNNs) solely to AIS trajectories for ship behavior classification and anomaly detection is currently on the rise (Li et al., 2021; Ma et al., 2025). These studies effectively

Table 6. Summary of the advantages and recent trends in multi-sensor fusion

Reference	Advantages of Multi-Sensor Fusion	Recent Developments and Trends
Rodger and Guida (2020), Zhao et al. (2014)	**Reduced observation gaps and failure modes** through complementary sensing by SAR (all-weather), optical imagery (visual identification), AIS (self-reporting), and RF sensing	SAR–AIS integrated surveillance frameworks have become a baseline architecture for wide-area maritime monitoring
Li et al. (2025a), Rodger and Guida (2020)	**Enhanced detection of dark and non-cooperative vessels**, where AIS-only approaches cannot distinguish intentional switch-off from reception loss	Fusion of SAR/optical detections with AIS absence using spatial, kinematic, and size-consistency constraints
Rodger and Guida (2020), Voinov et al. (2018)	**Reduction of false alarms and improved** detection reliability by cross-validating independent sensor observations	Structured fusion pipelines combining detection, feature extraction, matching, and verification across sensors
Habtemariam et al. (2015)	**Improved track continuity** by compensating for AIS coverage gaps and satellite revisit limitations	Measurement-level fusion of AIS and radar/SAR data for consistent multi-target tracking
Guerriero et al. (2008)	**Support for operational decision-making**, including sensor prioritization, re-tasking, and confidence-aware alerting	Performance- and information-value–driven fusion strategies linked to adaptive SAR tasking
Pleskachevsky et al. (2022), Tings et al. (2019a)	**Explicit recognition of environmental effects on sensor reliability**, motivating environment-aware fusion	Increasing use of SAR-derived sea-state parameters (wind, wave, swell) as priors for detection and fusion
Kanjir et al. (2018)	**Clear division of sensor roles (Trigger–Confirm paradigm)**, improving operational efficiency	SAR used for wide-area triggering, optical imagery for confirmation and classification under favorable conditions
Sørensen et al. (2024)	**Extension toward triple- and multi-modal fusion**, incorporating RF sensing as an independent cue	Emerging RF–SAR–AIS fusion studies for complex or high-risk maritime environments

model complex time-series/spatio-temporal ship trajectories as non-Euclidean graph structures, achieving high classification accuracy. This work lays the foundation for future expansion into Graph-based Multimodal Models that incorporate physical sensor data (radar, cameras) and environmental information.

5.4.3. Transformer-Based Multimodal Fusion

SAR–optical–AIS joint learning frameworks have recently emerged. Ziling et al. (2024) introduced a SAR ship recognition network enhanced by optical and AIS auxiliary data, achieving improved classification in cluttered scenes. The dark ship detection research by Li et al. (2025a) integrates SAR, Optical, and AIS data to perform an improved multi-feature association method based on oriented bounding boxes, which provides an efficient technical tool for identifying non-cooperative vessels using multi-sensor information. Wang et al. (2025) proposed TransOSS, a transformer-based cross-modal alignment network for optical–SAR ship re-identification.

5.4.4. Integration of Environmental Information

Environmental information integration remains a developing area. Colin et al. (2022) employed DL models to semantically segment metocean processes directly from Sentinel-1 SAR, demonstrating the feasibility of extracting environmental fields from SAR imagery for downstream tasks. Additionally, Tings et al. (2019a)'s metocean-based detectability model provides a quantitative foundation for connecting environmental conditions with sensor performance. In summary, instances of sea-state-aware Transformer or GNN architectures that concurrently utilize MetOcean, SAR, AIS, and RF as inputs are highly restricted. The majority of current efforts are focused on 2–3 modality combinations, such as (1) Imagery + AIS, (2) Imagery + Radar, or (3) AIS + Imagery/GPS. Nevertheless, multimodal AI frameworks that explicitly incorporate environmental data—including sea state, wind, waves, and currents—remain at an early stage, constituting a significant research gap for future exploration. Table 6 presents and summarizes the advantages and recent trends of multi-sensor fusion studies, categorized by research theme.

6. Discussion and Future Directions

6.1. Environmental Characteristics of Korean Waters

The Korean Peninsula is located in a region where bathymetry, sea-state dynamics, and atmospheric forcing vary sharply across the Yellow Sea, East Sea, and South Sea. These pronounced environmental contrasts exert a strong influence on SAR backscatter characteristics, optical visibility, AIS reporting reliability, and, consequently, the overall performance of multi-sensor MDA systems.

In the Yellow Sea, the combination of shallow bathymetry, strong tidal mixing, and high suspended sediment concentrations creates a particularly challenging detection environment. Elevated turbidity significantly reduces optical visibility while simultaneously increasing variability in SAR backscatter due to enhanced surface roughness and sediment-related scattering effects. In addition, the Yellow Sea is characterized by extremely dense vessel traffic, leading to frequent AIS message congestion and time-slot collisions, which in turn degrade track continuity and complicate the interpretation of vessel behavior. The East Sea presents a contrasting environmental regime dominated by deep bathymetry and strong stratification. Long-period swell events originating from the open ocean are a defining feature of this region and play a critical role in modulating SAR clutter statistics. These swell systems can introduce large-scale surface tilting and coherent streaks in SAR imagery, which distort wake signatures and complicate both vessel detection and classification, even under otherwise favorable imaging conditions. In the South Sea, complex coastal geometry interacts with strong monsoon forcing to produce highly dynamic environmental conditions. Rapid temporal changes in sea-surface roughness, frequent fog formation, and pronounced seasonal contrasts collectively reduce detection performance across sensing modalities. Increased surface roughness elevates SAR clutter levels, while persistent fog and cloud cover significantly lower the success rate of optical imaging. As a result, both SAR- and optical-based surveillance systems experience heightened uncertainty in this region.

6.2. Potential for Environment-Aware MDA in Korea

The integration of SAR, AIS, and environmental data provides a strong foundation for implementing environment-aware MDA in Korean waters. In particular, the fusion of KOMPSAT-5 X-band SAR with AIS information and MetOcean fields—including wind, wave conditions, turbidity, and surface currents—enables the development of a regionally optimized detection and matching framework that explicitly accounts for the environmental characteristics of the Korean EEZ. Such integration allows vessel detectability and sensor reliability to be interpreted in the context of prevailing sea-state conditions, reducing ambiguity in both detection and AIS association. Beyond single-sensor studies, multi-platform maritime monitoring frameworks have also been explored in domestic research. Kim et al. (2020b) integrated

satellite observations, Unmanned Aerial Vehicle (UAV) data, AIS, and HF radar to support scenario-based ship monitoring, illustrating the practical value of multi-sensor fusion for maritime domain awareness in Korean waters.

The growing availability of high-revisit commercial SAR constellations further enhances the feasibility of environment-aware surveillance. Platforms such as ICEYE and Capella support frequent monitoring of dynamic maritime environments, making it possible to incorporate near-real-time sea-state estimates into adaptive tasking strategies and detection thresholding. By conditioning acquisition planning and algorithmic parameters on evolving environmental conditions, these systems can improve detection robustness and reduce false alarms, particularly during periods of rapidly changing wind and wave regimes.

At the operational level, environment-aware MDA offers significant potential for integration across national agencies. Defense organizations, the Ministry of Oceans and Fisheries, and the Korea Institute of Ocean Science & Technology (KIOST) increasingly share common requirements related to EEZ monitoring, fisheries regulation, and dark-vessel detection. A multi-modal fusion framework that combines SAR, AIS, MetOcean, and RF information can support mission prioritization, enhance early-warning capabilities, and provide a scalable foundation for the development of a national MDA architecture tailored to Korea's complex and highly variable maritime environment.

6.3. Future Directions and Challenges

This section outlines the critical research directions and structural challenges that must be addressed to enable the reliable deployment of environment-aware surveillance technologies within MDA systems.

6.3.1. Lack of Integrated Environment-Aware Detection Models

Current SAR-, optical-, and AIS-based detection frameworks typically address environmental factors and sensor-geometry variables—such as sea state, surface slope, incidence angle, and AIS data gaps—in a fragmented and modality-specific manner. While these factors are often acknowledged individually, they are rarely integrated within a unified detection architecture. As a result, existing systems struggle to capture the coupled effects of environmental forcing and sensor response. There is therefore an urgent need to develop end-to-end detection models that holistically incorporate interacting environmental, geometric, and sensing variables into a single environment-aware framework.

6.3.2. Insufficient Real-Time Sea-State Adaptive Thresholding

Most operational maritime detection systems continue to rely on fixed or weakly adaptive thresholds, which are inherently ill-suited to non-stationary sea clutter conditions. Because sea clutter statistics vary dynamically with wind and wave conditions, static thresholding leads to unstable false-alarm rates and inconsistent detection performance. Future surveillance systems must incorporate mechanisms that enable real-time or near-real-time adjustment of detection thresholds and alarm criteria based on evolving sea-state conditions, thereby maintaining robustness across a wide range of environmental regimes.

6.3.3. Limited Multi-Sensor Datasets with Environmental Labels

The development of environment-aware machine learning models is severely constrained by the lack of multi-sensor datasets annotated with environmental metadata. Most publicly available SAR ship-detection datasets provide only image-level annotations, without accompanying information on wind speed, wave height, clutter intensity, or atmospheric conditions. This limitation restricts the ability of learning-based models to generalize across diverse environmental scenarios and hampers systematic evaluation under extreme sea states. Addressing this gap requires the creation of standardized, multi-sensor datasets that explicitly include sea-state and atmospheric labels.

6.3.4. Challenges in Small-Vessel Detection Under High Sea States

Small and low-radar-cross-section vessels remain particularly difficult to detect under rough-sea conditions, where intensified sea clutter significantly degrades target-to-clutter contrast. In such environments, both conventional detectors and deep learning–based approaches experience elevated false-alarm rates and increased missed detections. Overcoming this limitation necessitates the development of detection algorithms explicitly designed for high sea states, as well as dedicated benchmarks and evaluation protocols focused on rough-sea scenarios.

6.3.5. Absence of a National Roadmap for Korea's Environment-Aware MDA

Despite the operational importance of maritime surveillance in Korean waters, there is currently no unified national roadmap guiding the development and deployment of environment-aware MDA systems. The Korean maritime domain is characterized by shallow-water dynamics, strong tidal forcing, monsoon-driven variability, and complex coastal morphology, all of which impose distinctive challenges on satellite-based sensing. A coordinated

roadmap is therefore required to align sensor tasking strategies, data standardization efforts, and environment-aware surveillance operations specifically tailored to these regional conditions. Establishing such a roadmap represents a key strategic priority for advancing sustainable and operationally robust MDA capabilities in Korea.

7. Conclusions

This review demonstrates that environmental conditions are not peripheral background factors but fundamental constraints that shape sensor observations, detection algorithms, and behavioral inference within MDA. Variability in wind, wave height, swell, atmospheric state, and surface radiative properties directly alters SAR clutter statistics and wake/shadow visibility, governs the contrast and usability of optical imagery, and induces environment-driven AIS reporting gaps or darkening. These effects cascade through the entire sensing and fusion chain, ultimately determining the reliability and interpretability of multi-sensor maritime surveillance. The central implication is clear: MDA systems that do not explicitly incorporate environmental awareness are inherently limited in their ability to maintain stable performance in detection, tracking, and vessel characterization. Because environmental fields are spatiotemporally non-stationary—and because each sensor exhibits distinct failure modes under varying conditions—future architectures must move beyond sensor-specific enhancements. Instead, sensing, algorithmic design, and behavioral modeling need to be reframed with the environmental field as a core organizing principle.

This perspective carries particular operational significance for the Korean EEZ and the broader East Asian seas, where shallow-water turbidity, monsoon-induced variability, long-period swell, and dense vessel traffic create especially challenging surveillance environments. In such regions, environment-aware MDA offers substantial advantages: the fusion of KOMPSAT-5, high-revisit commercial SAR, satellite AIS, and MetOcean datasets can support adaptive detection strategies, condition-dependent reliability weighting, and systematic interpretation of cross-sensor inconsistencies. These capabilities form the basis for next-generation Korean maritime surveillance systems. In summary, environment-aware MDA should be regarded not as an optional enhancement but as a foundational requirement for future maritime monitoring and security. The framework and research directions outlined in this review are intended to support the development of robust, environmentally adaptive MDA solutions tailored to the Korean EEZ and other dynamically evolving maritime regions.

Author Contributions

Conceptualization: Ryu SM, Han SH; Data curation, Formal analysis: Ryu SM; Funding acquisition: Ryu SM, Han SH; Investigation, Methodology: Ryu SM; Supervision: Han SH; Validation, Writing–original draft, Writing–review & editing: Ryu SM.

Conflicts of Interest

No potential conflict of interest relevant to this article was reported.

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Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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Supplementary Materials

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